



Module Title
Cybersecurity and Networking

Internship Assignment Submission

Student Name: Jiwosh Shrestha

Internship ID: JVL-CS-2026-8990

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1.Question and Answers

1. What is a computer network?

A computer network is a group of computers and devices connected together so they can communicate and share information and resources, such as files, printers, and internet connections.

2. What is the difference between a client and a server?

A client is a device or program that requests a service or resource.

A server is a computer or program that provides the service or resource.

Example: When you open a website, your browser is the client, and the computer hosting the website is the server.

3. What is a LAN?

LAN (Local Area Network) is a network that connects devices within a small geographical area, such as a home, office, school, or college.

Example: Computers connected within a college computer lab.

4. What is a WAN?

WAN (Wide Area Network) connects networks over a large geographical area, such as between different cities or countries.

Example: The Internet is the largest WAN.

5. What is the purpose of a switch?

A switch connects multiple devices within the same network and sends data to the correct device.

Example: It can connect computers, printers, and servers in an office network.

6. What is the purpose of a router?

A router connects different networks and forwards data between them.

Example: A home router connects your local home network to the Internet.

7. What is a firewall?

A firewall is a security system that monitors and controls incoming and outgoing network traffic based on security rules.

Its main purpose is to protect a network or computer from unauthorized access and attacks.

8. What is an IP address?

An IP address (Internet Protocol address) is a unique address given to a device on a network.

It helps devices identify and communicate with each other.

Example: 192.168.1.10

9. What is a MAC address?

A MAC (Media Access Control) address is a unique physical address assigned to a device's network interface.

It is mainly used to identify devices within a local network.

10. What is a default gateway?

A default gateway is the device, usually a router, that allows devices in a local network to communicate with devices on other networks or the Internet.

Simple example: If your computer wants to access a website, it sends the traffic to the default gateway (router).

11. What is DNS?

DNS (Domain Name System) translates website names into IP addresses.

Example: Instead of remembering an IP address, you type a website name, and DNS finds the correct IP address for it.

Simple meaning: DNS is like the phonebook of the Internet.

12. Why is networking knowledge important for cybersecurity?

Networking knowledge is important for cybersecurity because most cyber-attacks happen through networks and network-connected systems.

Understanding networking helps cybersecurity professionals:

- Identify suspicious network traffic
- Configure firewalls
- Detect attacks
- Understand IP addresses and protocols
- Secure network devices
- Troubleshoot security problems

2. Network Information (Practical Lab)

13. Write your computer's private IPv4 address?

Ans: My computer's private IPV4 address is 192.168.101.4 255.255.255.0

14. Write your default gateway.

Ans: My computer's default gateway is 192.168.101.1

15. Write your DNS server.

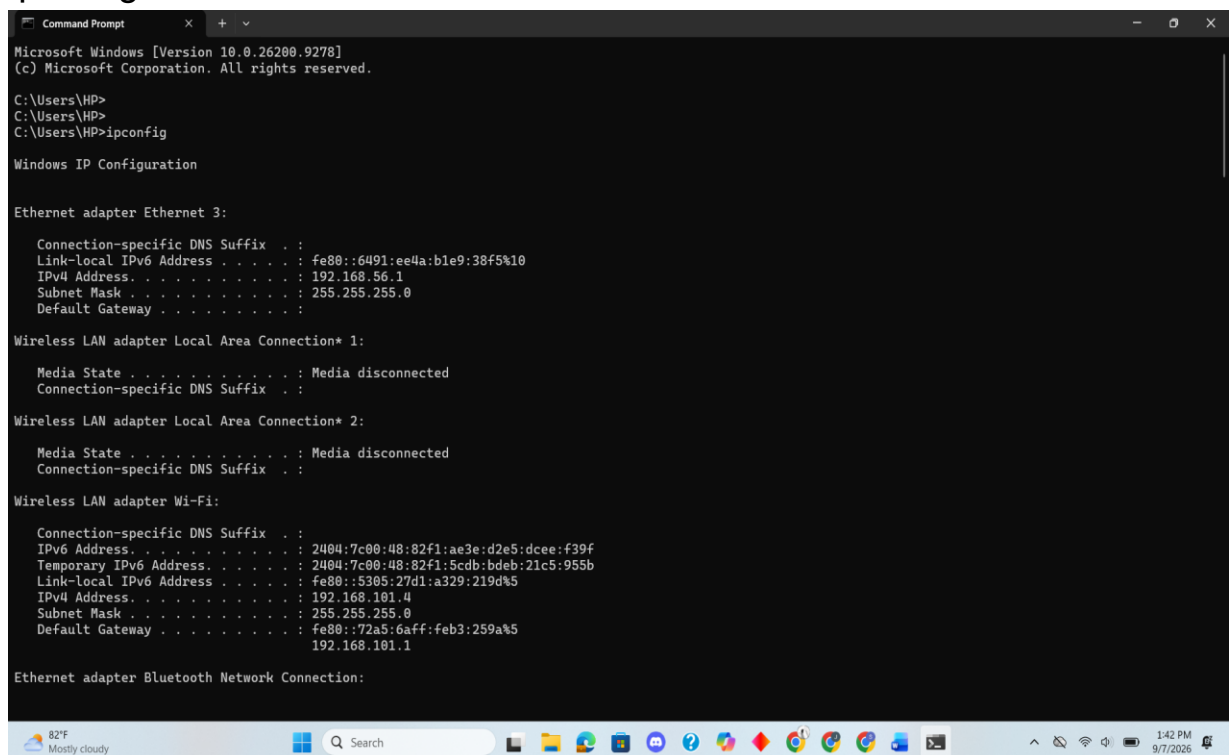
Ans:

Primary DNS: 110.44.112.245

Secondary DNS: 110.44.113.245

3. Attaching screenshots of: ipconfig, ping, tracert

➤ Ipconfig:

A screenshot of a Windows Command Prompt window. The title bar reads "Command Prompt". The text inside shows the command "ipconfig" being executed. The output displays network configuration for three adapters: Ethernet adapter Ethernet 3, Wireless LAN adapter Local Area Connection* 1, and Wireless LAN adapter Wi-Fi. The Ethernet adapter is connected and shows an IPv4 address of 192.168.56.1. The two wireless adapters are disconnected. The taskbar at the bottom shows the date and time as 1:42 PM on 9/7/2026.

```
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>
C:\Users\HP>
C:\Users\HP>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::6491:ee4a:b1e9:38f5%10
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2404:7c00:48:82f1:ae3e:d2e5:dcee:f39f
    Temporary IPv6 Address . . . . . : 2404:7c00:48:82f1:5cdb:bdeb:21c5:955b
    Link-local IPv6 Address . . . . . : fe80::5305:27d1:a329:219d%5
    IPv4 Address. . . . . : 192.168.101.4
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::72a5:6aff:feb3:259a%5
                                192.168.101.1

Ethernet adapter Bluetooth Network Connection:
```

➤ Ping:

```
C:\Users\HP>ping google.com

Pinging google.com [2404:6800:4007:83e::200e] with 32 bytes of data:
Reply from 2404:6800:4007:83e::200e: time=128ms
Reply from 2404:6800:4007:83e::200e: time=134ms
Reply from 2404:6800:4007:83e::200e: time=107ms
Reply from 2404:6800:4007:83e::200e: time=153ms

Ping statistics for 2404:6800:4007:83e::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 107ms, Maximum = 153ms, Average = 130ms
```

➤ Tracert:

```
C:\Users\HP>tracert google.com

Tracing route to google.com [2404:6800:4007:83e::200e]
over a maximum of 30 hops:

  1      8 ms      1 ms      1 ms  2404:7c00:40:5::cc:5d6e
  2     10 ms      3 ms      3 ms  2404:7c00::12
  3      9 ms      2 ms      4 ms  2404:7c00:0:d::1
  4      4 ms      3 ms      7 ms  2404:7c00:0:4007::1
  5     10 ms      4 ms      4 ms  2404:7c00:0:4002::1
  6     13 ms      7 ms      6 ms  2404:7c00:0:3::b
  7     14 ms      8 ms      7 ms  2404:a800:1a00:807::89
  8    127 ms     73 ms     124 ms  2404:a800::92
  9      *        *        *      Request timed out.
 10   143 ms    101 ms    100 ms  2404:6800:8201:300::1
 11   105 ms    201 ms     49 ms  2001:4860:0:1::464
 12    67 ms     99 ms    102 ms  2001:4860:0:1::5595
 13    99 ms    101 ms    101 ms  lcmaaa-bc-in-x0e.1e100.net [2404:6800:4007:83e::200e]

Trace complete.
```